



How can I return multiple values from a function?

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We all know that a function in C can return only one value. So how do we achieve the purpose of returning multiple values.

Well, first take a look at the declaration of a function.

```
int foo(int arg1, int arg2);
```

So we can notice here that our interface to the function is through arguments and return value only. (Unless we talk about modifying the globals inside the function)

Let us take a deeper look...Even though a function can return only one value but that value can be of pointer type. That's correct, now you're speculating right!

We can declare the function such that, it returns a structure type user defined variable or a pointer to it . And by the property of a structure, we know that a structure in C can hold multiple values of asymmetrical types (i.e. one int variable, four char variables, two float variables and so on...)

If we want the function to return multiple values of same data types, we could return the pointer to array of that data types.

We can also make the function return multiple values by using the arguments of the function. How? By providing the pointers as arguments.

Usually, when a function needs to return several values, we use one pointer in return instead of several pointers as arguments.

Please see [How to return multiple values from a function in C or C++?](#) for more details.

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